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Alphabet: Google 2030

02 December 2025

Alphabet

Rating: Buy

Ticker: GOOG US

Market Cap.: \$3.8tr

Price at 1 Dec '25: \$315

Target Price: \$380 (+21%)



1. We review *long-term* prospects for Alphabet to *materially* grow sales and profits, while recapping our prior work on cost savings, its numerous dominant franchises and 2bn+ user properties.
2. The main way is by *finally* improving *productisation* of world-class AI research/infrastructure, seen in Gemini3, Veo3, GCP RPO, and engagement with erstwhile rivals (Apple, Meta, etc.).
3. We think this will bring \$40bn in Google One (Gemini, Drive, APIs, etc.) subscriptions by 2030.
4. Cloud is reaching new plateaus in scale; we forecast \$250bn in 2030 sales (incl. \$18bn in TPUs).
5. YouTube remains a unique asset, the core of a Google bundle, and should get sharper disclosure. We forecast YouTube topping \$100bn+ in Ads, Subscriptions, TV, and Sunday Ticket by 2030.
6. Google keeps replacing opex with capex and growing its best-in-class E2E DC infrastructure. It avoided major regulatory actions impacting its earnings, with ad tech remedies likely to be mild.
7. The P&L still burdened by 12% of sales at *low margins* or a *loss* (Network at 9%, Hardware at 3%). Addressing these is an opportunity for further margin accretion, in our view.
8. The company has a long list of “option values” – Waymo, Verily, Isomorphic Labs, Quantum, etc. We think Google *could* sell \$18bn of TPUs by 2030, seeding 3P/NEO clouds, limiting its own capex.
9. We see Google nearly doubling sales to over \$700bn in 2030E, as its mix shifts to high-growth units (Cloud, YouTube, AI subscriptions/monetisation), with margins rising to 41% (despite higher D&A).

Key Insights

- We have had a Buy rating on Alphabet for several years, **expecting the company to resolve perceived or real shortcomings in productising its deep AI research.**
- Google also has **exceeded our – and market – expectations with the growth and profits of its Cloud business.**
- **This should not preclude further structural changes to improve the business:** Google has 12% of sales in low-margin or loss-making businesses with limited potential, in our view.
- Google is **already at record margins**, applying capex to replace opex and **using AI to realise internal efficiencies.**
- **We keep our Buy rating, introduce forecasts to 2030** (and a view of what the company might look like by then), and make the argument that, like Apple, the **duration in Alphabet's business model deserves a premium 30x P/E valuation.**
- **Our YE'26E price target rises from \$300 to \$380.**

In '26, we expect the following catalysts:

Further **extension of Gemini distribution** – reaching 1bn+ users in 1H26E, 2bn by YE'26E – and **further examples of monetising leading AI Research.**

Accelerating Cloud growth rates (based on RPO, adding Wiz to sales, leveraging infrastructure investment), **topping \$250bn of 2030E revenues.**

Unified YouTube reporting, potentially with EBIT disclosure (as helped with Cloud). **On track for \$100bn+ in 2030E sales as #1 in global video.**

Lower FCF margins (assuming capex of \$124bn and \$149bn in '26E and '27E), **but supporting 14% sales growth CAGR to 2030E.**

Table 1: Summary Financials, '24-'27E

Y/E to Dec.	Sales	Growth	GAAP EBIT	EBITDA	GAAP EBIT Margin	GAAP EPS	GAAP P/E	EV/ EBITDA	FCF Yield
2024	\$350bn	14%	\$112bn	\$128bn	32.1%	\$8.0	39.2x	29.6x	1.9%
2025E	\$400bn	14%	\$133bn	\$157bn	33.2%	\$10.7	29.4x	24.3x	1.6%
2026E	\$458bn	14%	\$159bn	\$192bn	34.7%	\$11.3	27.8x	19.7x	1.8%
2027E	\$521bn	14%	\$186bn	\$235bn	35.8%	\$13.2	23.9x	16.1x	2.3%

Source: Company data, Arete Research estimates

Significant scope for management to add value from further structural changes.

From Under- to Out-Performance

Starting Point:

- Our [Time for a Shake Up](#) (May '23) note came as Google **underperformed**, despite \$70bn buybacks.
- Perception was of **failure to productise a decade of AI investment** and sclerotic/uninspiring leadership.
- That was reversed by 1) quashing the “Search is Dead” narrative; 2) executing on the **AI product roadmap**; and 3) growing ancillary (non-Ads) businesses.

Fig. 1: Alphabet Under-, Then Out-Performed Nasdaq Since June '22



Source: FactSet

Current Actions:

Distribution: Extending Reach

- Google addressing “Android problem” (loss of market share) via Siri model deals with Apple. (see [Arete Digital Download: April '25 Edition](#) – “Grand Bargain” (Apr. '25).
- Retains **vast reach** with **ten 2bn+ user properties**.

Cost: Partly Addressed, Shifting to Capex

- Headcount/Employee Comp. flat, as capex related opex ([Spring Cleaning](#), Feb. '23).
- Google spent \$82bn in opex in '22, rising \$18bn to \$100bn in '25E;
- During same period, sales rose by \$118bn.

Key Questions:

- What incremental revenue streams can Google monetise?
- How does its AI investment (esp. capex) impact ROIC/ROCE?
- Can it address low-growth/loss-making efforts, partly in “run-off” mode?
- Cloud – What next given meteoric rise in RPO, reaching 20%+ margins?
- YouTube – #1 *global* video brand, in transition from **Ads to #1 global Subs base**?
- Ads and Search – Aiming for transition, ready for 10-100x rise in Search query volume.

Google benefits from singular AI focus, but has multiple option values to realise

Google 2.0 Note (Apr. '19): Next \$100bn

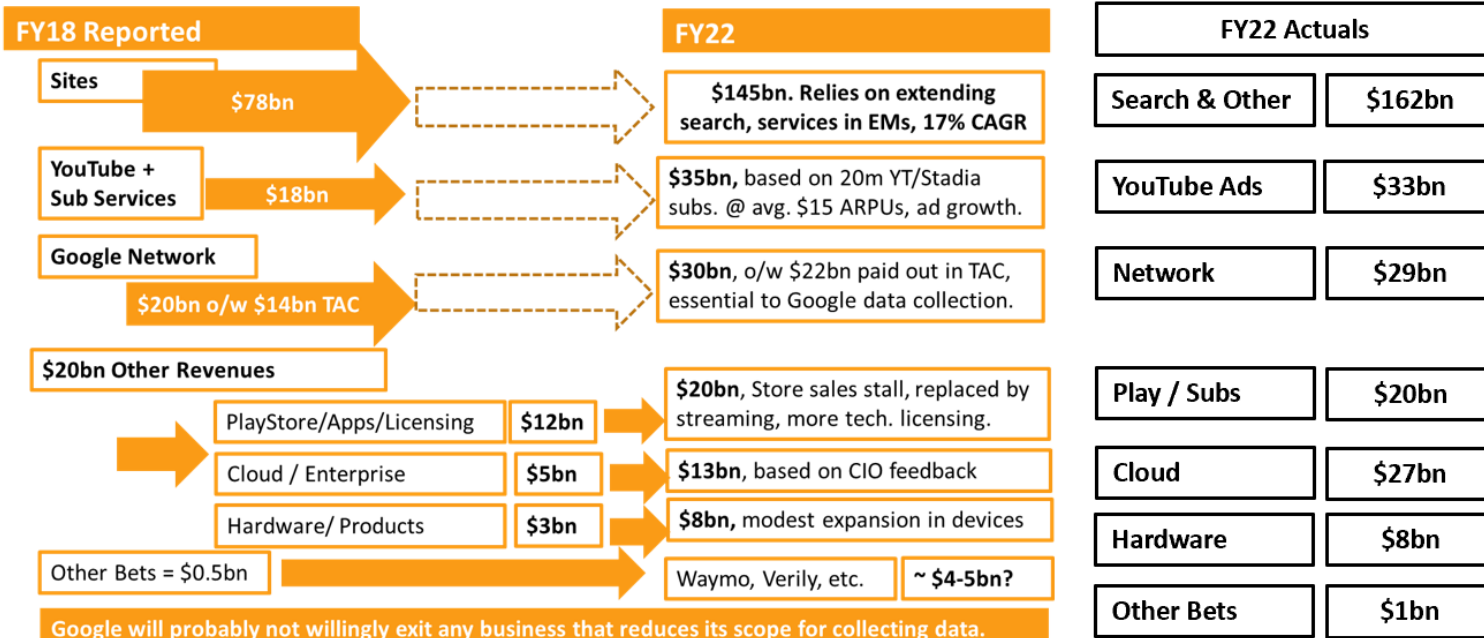


Future Google: Finding the Next \$100bn

10 April 2019

One Hundred Billion Dollars

Fig. 1: From Arete's 2019 "Future Google" Report



Beyond "puts & takes" (higher Cloud, no Stadia), direction from 2018 was clear.

Google 3.0 ('23): Time for Shake-Up



Create a Valid Consumer Bundle

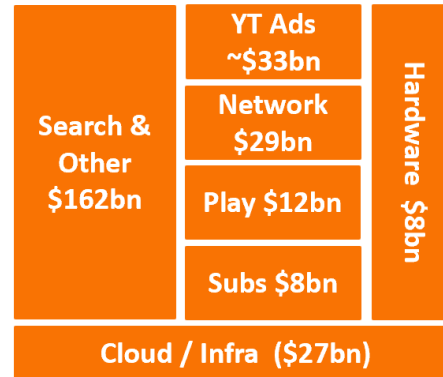
Create an AI “Product Factory” [incl. Search]

Reboot Hardware

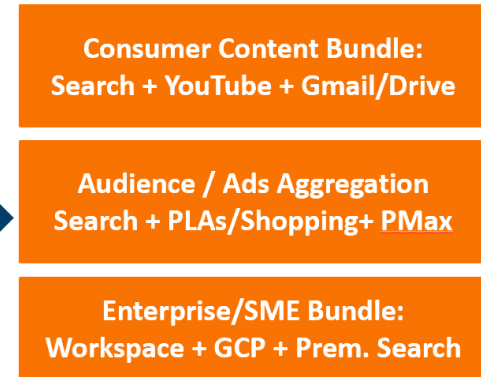
Short Ad Tech:
Spin-Out 3P Network

De-Couple
Cloud/YouTube

From:



To:



Consumer Bundle: YouTube Music + Premium + Smart Gmail (with AI) + Drive + VPN + Voice + Pay + Fitness? Options for TV/Sunday Ticket. (~100m YouTube subs / \$11bn FY23E sales).

Enterprise Bundle: Workspace, Pre-Trained Models, AlaaS / APIs; + access to compute/storage + cybersecurity. Premium AI tools bundled with GCP capacity.

Source: Arete Research.

- Google managed to recover share price momentum by **ONLY** addressing **ONE** of these **issues** (productising its AI research).
- Aided by “incremental” growth in Cloud and a recovery in YouTube Ads growth.
- Still carrying ~12% of sales which are low margin and shrinking (Network) or losing money (Hardware) alongside Other Bets scaling.

Google 2030: Offense vs. Defense

Defense:

- **Prevent loss of query volume share in Search** to OAI, other LLMs.
- Capture more **discretion over ad spend** with PMax.
- Aggressively scale ever-lower token costs.

From...

- **Ads being 73% of 2025E sales...**
- **Cloud at 14% of sales...**
- **Subscriptions rising from 6% of sales...**

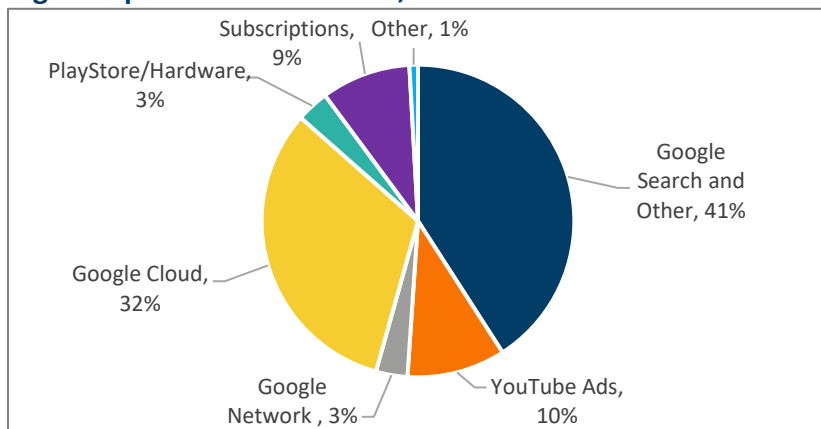
Offense:

- **Capture other “TAMs” with *vertical* AI applications** (Isomorphic Labs, Verily, many GCP customers).
- **Apply AI research to adjacent markets** (pharma, robotics, energy, public sector, etc.)

To...

- **Ads being 54% of 2030E sales** (S&O, YT, Network).
- **Cloud reaching 32% of sales.**
- **Subs at 10%, incl. Gemini and YouTube/bundles.**

Fig. 2: Alphabet Business Mix, 2030E



Source: Arete Research estimates

- We see a “fuzzy line” between Cloud sales, Subscriptions (Gemini, YouTube) and Search & Other by 2030.
- All business units likely **powered by Google AI models and run on GCP E2E infrastructure**
- **Hard to measure efficiency gains in next generations of TPUs, inference, agents, further automation, etc.**

Search is not dead. It is changing. AI impacting all parts of Google.

Margins by Division

Table 2: Alphabet Margins by Division

	'24	'25E	'26E	'27E
Search	92,716	106,550	118,457	131,005
Network	917	1,713	1,797	1,507
YouTubeAds	8,314	11,304	13,995	16,783
YouTube Subs	687	1,268	2,476	3,944
Devices	-2,483	-2,484	-2,106	-1,852
PlayStore	8,947	9,741	9,357	9,496
Cloud	6,119	12,709	21,054	31,462
OtherBets+Lic./Pay, etc	-2,990	-3,413	-4,979	-4,662
Total EBIT	112,226	137,387	160,052	187,682
Act/F'cast EBIT	112,390	132,707	159,110	186,460
Search	47%	49%	49%	50%
Network	3%	6%	6%	6%
YouTubeAds	23%	28%	29%	31%
YouTube Subs	4%	5%	8%	10%
Devices	-25%	-24%	-20%	-17%
PlayStore	77%	80%	80%	80%
Cloud	14%	22%	25%	27%
OtherBets+Lic./Pay, etc	-76%	-90%	-114%	-93%

Source: Company data, Arete Research estimates

'25E figures diverge from Forecast/Reported due to \$1.4bn (Texas) and \$3.9bn (EU) legal settlement costs.

Other Bets losses peak as Waymo rolls out in 14 cities by YE'26, before starting to generate material sales.

- Our estimates of margins by division, subject to opaque corporate cost allocation.
- Search goes from 80%+ of EBIT in '25E to ~70% in '27E and sub-65% in '30E.
- Cloud goes from 5% of group EBIT in '25E to 16% in '27E and ~25% by '30E.
- Google still "carries" 12% of sales in low-margin Network & loss-making Hardware, diluting margins.

This highlights urgency of monetising AI investments, defending Search position.

AI: Widest Agenda in Industry

Starting Point:

- Google has out-invested rivals in AI R&D for past decade.
- \$230bn+ capex since '21 (excl. real estate deals).
- AI at the core of its business ("AI-first" since 2016)

- Initial concerns about LLM commoditisation ([link](#)) shifted to inference/efficiency of models, tokenomics.
- **Gemini3 keeping pace with rival LLMs from OAI, ANTH**
- We see Google having a major **distribution** and **training dataset** advantage (Android/Chrome, YouTube, Maps, Gmail all 2bn+ users), while Google gets daily flow of O&O training data.

Core AI Products:

- Gemini family of models
- PMax
- AI Overviews/AI Mode
- Maps (supporting Waymo)
- YouTube/Veo3/Nano Banana
- Gmail/Workspace
- Cloud Optimisation

83K Engineering Staff (LinkedIn)

Vast Google Research Agenda

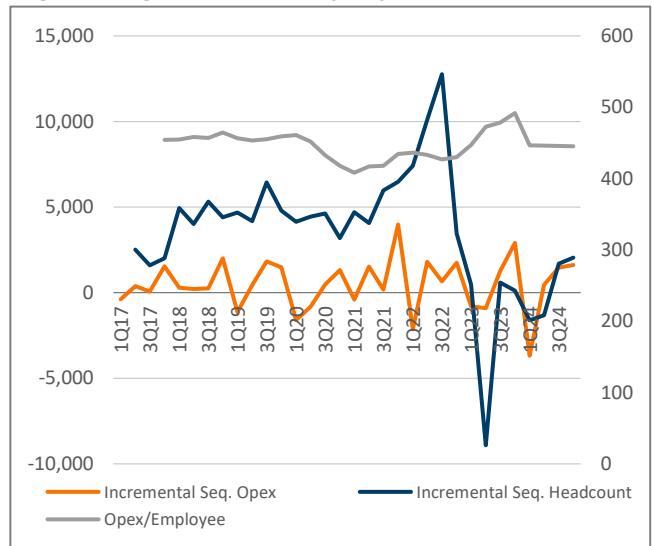
Foundational ML & Algorithms	Computing Systems & Quantum AI	Science, AI & Society
Algorithms & Theory	Distributed Systems & Parallel Computing	Climate & Sustainability
Data Management	Hardware & Architecture	Economics & Electronic Commerce
Data Mining & Modeling	Mobile Systems	Education Innovation
Information Retrieval & the Web	Networking	General Science
Machine Intelligence	Quantum Computing	Health & Bioscience
Machine Perception	Robotics	Human-Computer Interaction and Visualization
Machine Translation	Security, Privacy, & Abuse Prevention	Responsible AI
Natural Language Processing	Software Engineering	
Speech Processing	Software Systems	

Incl. future areas like Quantum, Computer Vision, Robotics, Science and AI

Roadmap Since I/O:

- Gemini 2.5 and 3
- AI Overviews/AI Mode
- Veo3/Nano Banana
- Aluminium OS (Chrome+ Android)
- AVs/Robotics

Fig. 3: Google TTM Sales/Opex per Headcount



Source: Alphabet filings

Alphabet has *demonstrated* AI can make its *own* operations incredibly efficient.

Search: Perception vs Reality

- From “Search is Dead” to Search is evolving to deeper “answers”.
- **SEO is not dead!** Vast industry behind \$225bn of Search & Other spend in '25E.
- SEO shifting to GEO (Gen AI Optimisation), Traffic Diversification.
- **AI Overviews/Mode** are *swiftly* changing Open Web dynamics.
- **Agents/AI can search 10-100x humans:** It's about monetising traffic, at lowest infrastructure cost (focus of years of Google tech. infrastructure effort).
- **Google has pricing power.** Indexing 2020 levels to 100, by '25E CPC has grown +34%, and Paid Clicks by +58%.

Table 3: Google Reported Paid Clicks Growth + CPC Matches Closely to Reported Search & Other Growth

	1Q23	2Q23	3Q23	FY23	1Q24	2Q24	3Q24	FY24	1Q25	2Q25	3Q25
Paid Clicks	8%	8%	7%	7%	5%	6%	4%	5%	2%	4%	4%
CPC Growth	-7%	-3%	4%	1%	8%	7%	8%	7%	7%	8%	8%
PC+ CPC	1%	5%	11%	8%	13%	13%	12%	12%	9%	12%	12%
Search & Other	1.9%	4.8%	11.3%	12.7%	14.4%	13.8%	12.2%	12.5%	9.8%	11.7%	14.5%

Source: Alphabet 10-Q/10-Ks

Google Ads begin surfacing inside AI Mode as tests expand

Source: Search Engine Land

Table 4: Search & Other CAGR

'20-'25E	16%
'25E-'30E	6%

Source: Arete Research estimates

- **Multiple *practical* moats around Google Search spend** (SEO firms, familiarity, SMEs devoted to it, still 100x or greater volumes than ChatGPT queries).
- **Search behaviour in transition** to higher-volume, greater reliance on quality scored/page ranked results.

Google's “Edge”

1. Cost per Query
2. Fidelity/Quality
3. Distribution
4. Ancillary Services
5. Brand
6. SEO Familiarity

AI Sparks Incremental Queries

ChatGPT Is Not Replacing Google —It's Expanding Search



Source: Semrush.

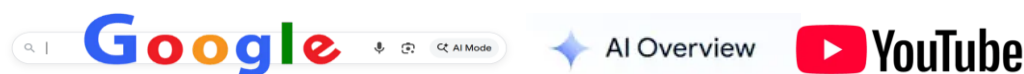
We assume a sharp slowdown in Search & Other; lower query volume, rising pricing

Distribution Wins

Google has 10 (!) services claiming 2bn+ users. They are #4 global brand (Interbrand), with YouTube at #13. Search, Android, Chrome, Gmail, Workspace, YouTube, PlayStore, Drive (storage), Maps, and AI Overviews
 Google still has “holes” (e.g. consumer messaging, social properties outside YouTube, poor Pay penetration)
 Multiple ways to *unite* ecosystem in new segments (SmartHome, AVs/Waymo, Agents across properties).

Foundational Utilities:

Search/AI Overview/YouTube



OS Layer/Utilities:

Chrome/Android/PlayStore/ Maps



Productivity/Messaging:

Drive/Gmail/Workspace, + Docs, Sheets, Slides, Calendar w/ 500m+



Additional Services w/ 1bn+ users (or on track to reach that)



We see three key advantages of this ecosystem of services:

Massive datasets for training, inference and to include in products.
 Ex. **Veo3** trains on YouTube library.

Rapid time-to-market to launch new features/services
 Ex. AI Overviews hit 2bn in <1Y

API Linkages *Between* Services.
 Creates flywheel of usage. Still many lagging properties (e.g., Google Pay).

Google looks to have the most complete services ecosystem on a standard data model.

Cloud: Milestones in Sight

Starting Point:

- AWS took four years to go from \$25bn to \$80bn in sales.
- **Google Cloud on track to *more than double* from '23 to '26E** – going from \$33bn to \$80bn.
- We forecast \$245bn in 2030E, as wider Public Cloud reaches \$2tr by 2035E ([GenAI: The Fast and The Furious](#), June '25)
- G-Suite – 10-11m Ent. customers, growing slower than GCP.
- **Ramping far faster than anyone forecast.**

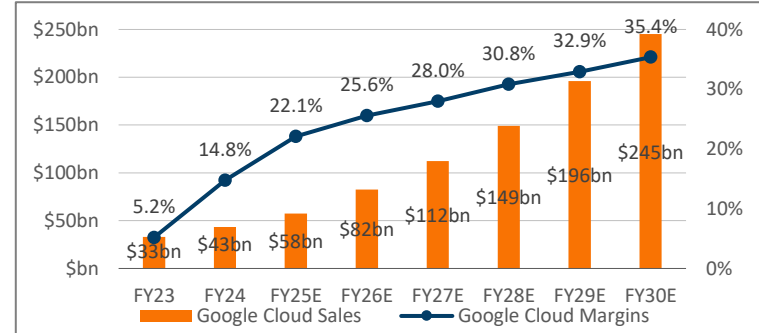
- **Clear operating leverage – 5% margins in '23, 15% in '24, and on track for 22% in FY25E.**
- GCP forecast \$12.7bn profits, \$58bn in FY25E sales, +33%.
- Some “inorganic” growth in '26E, adding \$1bn+ of sales in \$32bn Wiz multi-cloud security acquisition.
- **Leading position in cybersecurity**, with multiple prior VC investments (CrowdStrike, Zscaler, Chronicle).
- **We forecast 33% margin in 2030E, still below AWS peaks.**
- **We assume TPU sales of \$18bn in Cloud by 2030E.**

Cyber M&A: We anticipated acquisition of Wiz in '23 (@\$2bn!); we see **Astrix** or **Cyera** as next potential targets.

Surge in Backlog: RPO went from \$93bn at YE24 to \$158bn at 9M25, claims 13 products @ \$1bn+ annual sales.

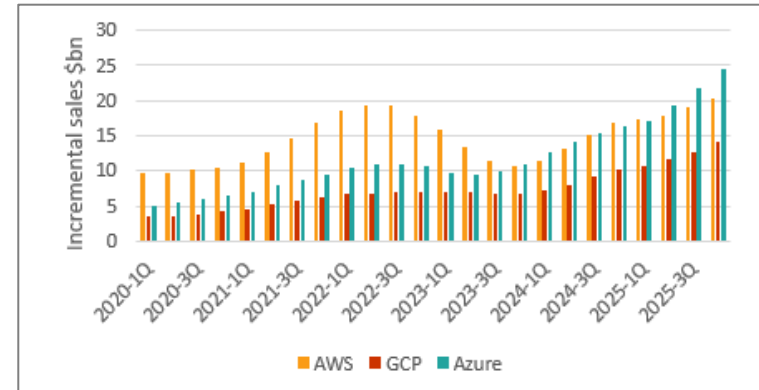
Google Cloud has clear momentum in sales, profits, and captured key AI customers.

Fig. 4: Cloud Growth



Source: Arete Research estimates.

Fig. 5: Incremental GCP Sales Coming Closer to AWS/Azure



Source: Company reports, Arete Research.

Nearly all

AI unicorns
use Google Cloud

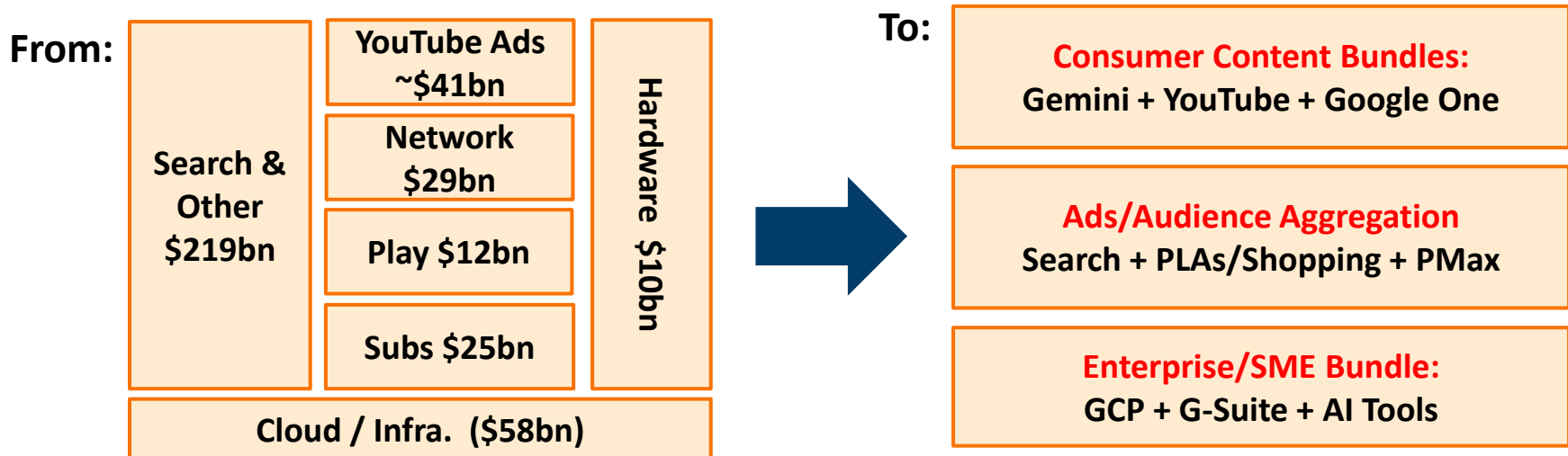
9

of the top 10 AI Labs
use Google Cloud

Source: Google Cloud presentation.

Consumer Bundles

- Longstanding issue with Google is **sub-optimal “matrix” org structure**. Failure of *many* projects ([link](#)).
- Ample scope to **simplify the offering for distinct audiences/TAMs**, improve capital allocation.
- Need more “customer-centric” approach within distinctly engineering-led culture.



Where is Google’s “all the best digital services for one low monthly price”?

Consumer: YouTube Music/TV, Premium, “Smart Gmail,” Drive/VPN Pay, Sunday Ticket, etc.

Enterprise Bundle: G-Suite, Gemini, AlaaS / APIs. GCP + cybersec. Premium AI tools bundled with GCP.

Google needs *simple* Consumer/Enterprise bundles: *still* makes enormous sense to us.

YouTube: TV or Not TV?

- **#1 global video platform, #13 global brand**
- YouTube Ads = **\$41bn sales ('25E)**
- YouTube Subs = **~\$20bn ('25E)**
- 100m+ subs/trials YouTube **Music/Premium/TV.**
- **Confusion** of separate reporting of Ads vs. Subs
- Incr. presence on large screen formats (e.g. NFL)

- **We expect Google to unify YouTube reporting in '26;**
 - YouTube Ads/Subs Sales and EBIT;
 - Incr. disclosure worked for Cloud.
- Increasingly **competing for TV dollars, e.g., 30sec non-skippable Ads, with 1P user data (from wider Google).**
- To grab TV budgets, **needs aggregated audience/reach.**
- **Sunday Ticket** shifts audience further onto large screens; we forecast \$4bn+ by FY27E sales (Ads + Subs).
- **Bundling:** Packaging *in-game, on-demand* NFL content.

NETFLIX

22.7x '27E EV/EBIT
8.9x '27E EV/Sales



Applied to '27E Ads +
Subs EBIT and Sales =
avg. of **\$575bn**

Subs. Higher Gross Profit vs. Ads

- Google says avg. gross margin dollars from Subscribers is higher than Ads users.
- Subs are gross margin accretive.

Other options/programmes:

YT UX 3.0: AI-edited documentaries

Expand livestreaming verticals (e.g. in education, tutorials)

Ramp Shorts monetisation

Dominates *niche* sports content



YouTube is already #1 global video service; AI can create personalised UX/content.

Apple: Strange Bedfellows?

Far greater alignment between Apple and Google than often understood:

- Annual “CEO Check-Ins” to “increasing revenue performance of each party”.
- Apple plays the role of gatekeeper (see [Apple: Gatekeeping](#), Nov. 25) for Gemini distribution; extends revenue share payments for search to integrating AI Overviews/Mode into a revamped Siri.
- The reported “Grand Bargain” slightly different than we imagined: Bloomberg cited a ~\$1bn payment from Apple to Google, which may explain Apple’s \$1.5bn seq. jump in R&D guide for the Dec. Q.
- Apple would “buy” a customised 1.2tr parameter model from Google to power a new Siri for “1H26 release”.
- **Apple won’t want Siri to hallucinate; it needs RAG** (Retrieval Automated Grounding) – i.e., a source of truth. **Apple needs an expedient go-to-market for a new Siri.**
- Gemini has access to vast pools of ranked/quality-scored data (Google search corpus), constantly refreshed.

Money Flow:	 → 	 → 
Now	Google pays Apple 40% rev. share on Safari Search traffic (now non-exclusive). Reported in DoJ trial to be \$20bn.	Apple is a top three customer of GCP “Bigfoot” project for iCloud storage. Apple Intelligence trained on TPUs.
Future	Google pays Apple rev. share on AI Overviews/AI Mode queries, and for Gemini distribution/subs via AppStore	Apple <i>remains</i> top three GCP customer; Apple buys custom Gemini models for Siri, uses Google for audience extension

One amusing aside: The CEO of Google anti-ageing venture Calico is *also* Apple’s Chairman of the Board (Art Levinson).

Win-Win: Apple gets tech for new Siri; Google sits as Apple’s *default* AI model partner.

Stragglers (1): Network Declines

- Network has seen **13 consecutive Qs** of declining absolute sales: this is **NOT** accidental, in our view.
- This is **9% of sales** but less than **1% of profits**: dilutive to margins and of dwindling strategic value.
- In a tightly managed company like Google with “OKRs” and “KPIs”, the slide in Network sales is likely a deliberate attempt to shift traffic from Open Web to O&O properties carrying no TAC (traffic acquisition costs).

Table 5: Year-on-Year Change in Absolute Sales at Google’s Ads Divisions: Network Decline Is Likely Not Accidental

\$m	1Q22	2Q22	3Q22	4Q22	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25
Search & Other	7,739	4,844	1,613	-697	741	1,939	4,487	5,416	5,797	5,881	5,359	6,014	4,546	5,681	7,182
YouTube Ads	864	338	-134	-670	-176	325	881	1,237	1,397	998	969	1,273	837	1,133	1,340
Network	1,374	662	-127	-830	-678	-409	-203	-178	-83	-406	-121	-343	-157	-90	-194

Source: Company reports, Arete Research.

We proposed Google Spin out 3P Network business, removing what CEO called “low-margin business” ([link](#)) back in 2020 testimony.

- There may be a *de facto* exit depending on Brinkema remedies due to be delivered in early '26.
- Google could still sell partner inventory(e.g., PINS, RDDT) on PG/PMP basis, keeping some tools for audience extension.
- A standalone Network business could **use AI tools to compete at far lower take-rates**; already *at scale/more efficient*.

Gross Revenue Comps ('25E):

GOOG DV360 (GOOG DV360 excl. Pmax)	\$29bn (\$19bn)
TTD	\$13bn
Criteo	\$4bn
AMZN 3P Ads (DSP/TAM/etc..)	~\$5bn
META (Audience Network)	~\$3bn

Source: Company reports, Arete Research estimates.

- Google is *materially* larger than rival SSPs (MGNI, PUBM, AdForm, Index) as measured by sales.
- AMZN set a precedent of mid-single digit take-rates for PG and high-single digit for PMP: **DV360 could follow this**.

Google has *chance to reset* low-margin Network unit in '26 post-antitrust remedies.

Stragglers (2): Pixel/Hardware

- Google has an “Android problem” despite *years* of effort with Pixel and OEM partners.
- Android vendors spend \$10bn p.a. on marketing to take share from each other, while losing share to Apple.
- We estimate Google loses \$2.5bn+ from Pixel/hardware, excluding G-Series chipset efforts.
- Other Google hardware efforts fragmented, winding down, lacking common brand. No big successes.

No common branding,
UX, design look/feel



Android Smartphone Sales, '25E

Samsung	\$69.3bn
Xiaomi (ex-China)	\$16.4bn
Motorola	\$11.8bn
Pixel (est.)	\$7.9bn

Source: Visible Alpha cons. for Samsung, Lenovo, Xiaomi

We think Google needs:

- Tighter integration of OS and Silicon in future Android/Chrome (Aluminium OS)
- Closer partnerships with QCOM and Samsung (e.g., partners on Moohan XR project)
- Extensions of Android XR (Kering, Warby Parker as 3P licensees/fashion partners)
- Need to distribute Gemini at the *edge* (in addition to Apple Siri model deals)



SAMSUNG

- Samsung should be *lead* vendor for all Android product
- Long-time #1 in Android vols.**
- Galaxy AI = Gemini distribution
- Only brand globally vs. Apple**



- Aim = Gemini on smartphones
- Distribution for G-Series chip IP
- Google needs multi-device edge
- Dist. Cloud Edge (**DCE**) now on **GA**
- ADT partnership for Home



- Align releases of Snapdragon and Android features
- Merge “on-device” AI w/ Tensor IP
- Build silicon for edge compute.

Keep aims for Pixel in mind: create reference design to *distribute* Google services.

Hardware: Silicon Dreams

- Google bought Agnilux in 2010, ex-PA Semis team that left Apple post acquisition.
- Google Brain team realised in 2013 that they needed custom ASIC for Google Assistant.
- Google officially launched TPU at Google I/O '16, but was in production already in '15.
- Uses unique “Systolic Array” architecture, with dramatically lower memory requirements.
- Inter-Chip Interconnect (ICI) in TPU Pods has peak Bandwidth of 1.2 TB/s vs Blackwell NVLink 5 at 1.8 TB/s.
- For scale-out, Google uses Optical Circuit Switch (OCS), competing with Nvidia’s InfiniBand and Ethernet.
- More energy efficient for specific applications - Google is front-end designer (the actual RTL of the design) while Broadcom is the backend physical design partner.

TPU v5p/e



TPU 6 – Trillium
(vs. Hopper)



TPU 7 (E/P) – Ironwood
(vs. Blackwell)



TPU 8E/P
'26?

Assume 3m die of TPU7, 1.5m packages, 64 per rack, which cost \$1.25m =
\$25bn of TPU production in '26E
Vast majority (90%+) goes for internal use (i.e. into GCP, avail. for 1P/3P workloads).

Assume 2K racks sold to 3P customers in '26E (mostly Fluidstack w/ power from TeraWulf). This is ~\$2.5bn of the \$10bn cited by AVGO. We assume this rises to 10K racks at \$1.75m in 2030 = \$18bn in sales.

Neo clouds remove capex/lease costs from balance sheet, plus value of warrants.



FluidStack



TERAWULF

Google gets enormous *internal* value/cost edge from TPUs, limiting case for 3P sales.



- Clear market leader in AVs vs. many challengers.
- Reaching 10% ride-hailing share in SF, 7% in PHX.
- Plans to roll out 14+ cities globally by YE'26.
- Perceived threat to UBER/LYFT, 10m+ rides served.

CapitalG

- Late-stage growth fund, \$7bn AUM.
- Stakes: Stripe, Databricks, Gusto, Canva, AlphaSense, Monzo, Rippling, Collibra, many more.



- Precision medical incubator with 1,200+ staff.
- Multiple tools (Lightpath, Workbench) getting adoption in pharma, healthcare.
- Partnerships: GSK, Sanofi, Roche, Novartis, Pfizer, Santen, Otsuka, + others.



- Biotech/Ai drug discovery start-up.
- AlphaFold won DeepMind CEO a Nobel Prize.



- As of 3Q25, Alphabet had \$64bn of non-marketable equity securities on balance sheet.
- Multiple notable exits and fair value upward revisions: OI&E includes \$22bn of "gains on securities" in period to 9M25.



Quantum Compute

- *Decades* of investment in Labs.
- Working with D-Wave since '07.
- Own **dedicated silicon** (Willow).
- \$50bn cum. market cap in listed QC plays.



Plenty of sources of embedded value across the "Google Mafia" ecosystem.

Table 6: EBIT Margins Steadily Improving

	1Q23	2Q23	3Q23	4Q23	1Q24	2Q24	3Q24	4Q24	1Q25	2Q25	3Q25	1Y 1H Ppts. Incr	2Y 2H Ppts. Incr
Alphabet	25.0%	29.3%	27.8%	27.5%	31.6%	32.4%	32.3%	32.1%	33.9%	33.9%*	33.9%*	1.2%	6.1%

Source: Alphabet filings; *excl. Texas legal settlement and EU Fine accrual.

- **Key earnings driver has been *replacing opex with capex*.**
- **Lowest-margin units are also slowest growing (Network + Devices = 12% of sales).**

Table 7: Google Employee Comp Costs, Services and Cloud

	FY23	FY24	FY25E
Services Employee Comp.	\$46.5bn	\$44.6bn	\$44.5bn
As % of sales	17.0%	14.6%	13.1%
Cloud Employee Comp.	\$19.0bn	\$20.5bn	\$21.9bn
As % of sales	57.6%	47.5%	37.8%

Source: Alphabet filings, Arete Research estimates

- **Google investing in Cloud staff – added \$3bn of employee comp from '23 to '25E, while sales rose \$24bn.**
- **More mature Services saw staff costs fall by \$2bn while sales rose by \$52bn from '23 to '25E.**

- Beyond cost savings, staff reductions need to address bureaucracy and political/insular culture
- Management talks about “**durable**” cost savings but will not be more specific. Last major RIF was Jan. '23
- **Market generally rewards cost reductions despite risk of disrupting operations**

Value realised from *simplifying* Alphabet's businesses.

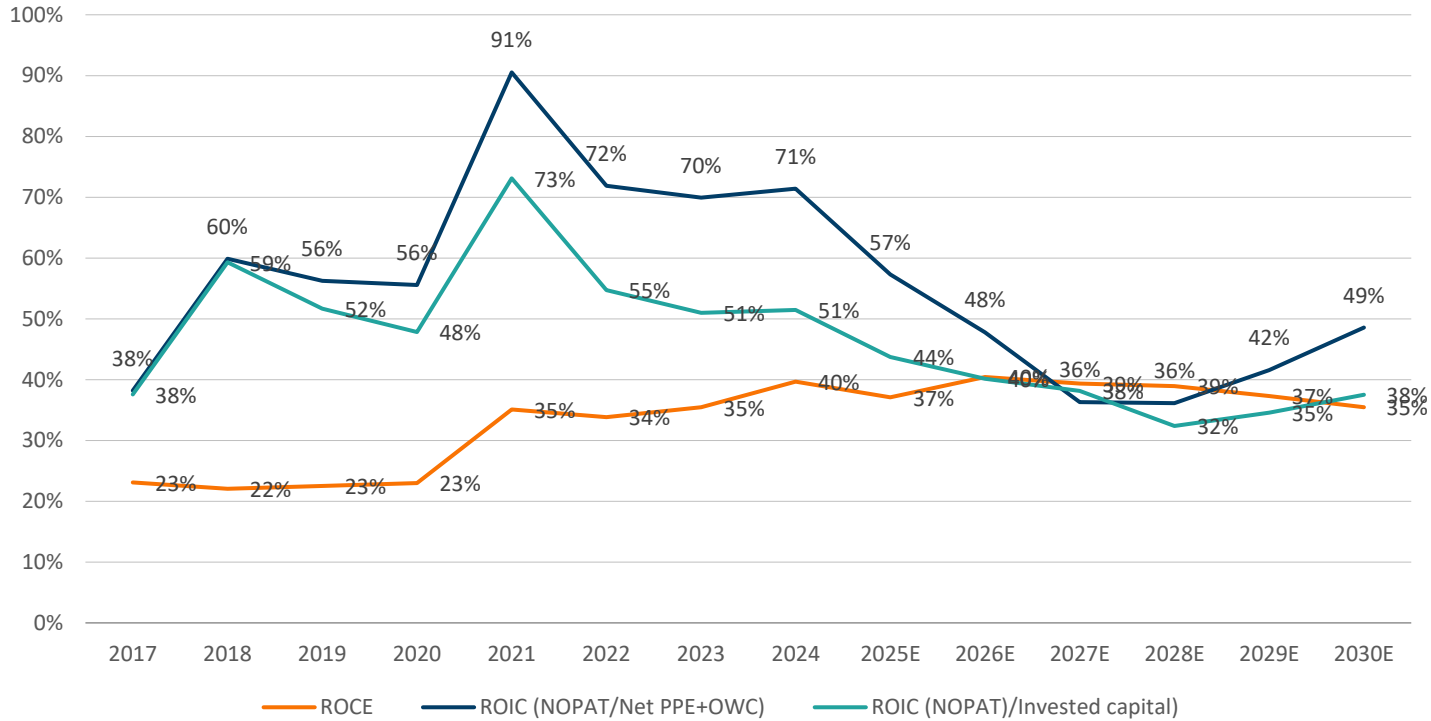
We see Google EBIT margins of 41% by 2030. led by mix, efficiencies and Subscriptions.

Returns vs. Investment Debate

ROCE and ROIC on Two Measures

- Google was making *impossibly high* returns in '21 (massive post-pandemic leverage).
- Capex surge will reduce these to still exceptional mid-30s levels in '26E-28E.
- Scope for rising returns as AI efforts mature and start to monetise at scale.

Fig. 6: ROCE and ROIC: 2017 to 2030E: Alphabet Heads Into an Investment Cycle



Source: Company data, Arete Research estimates

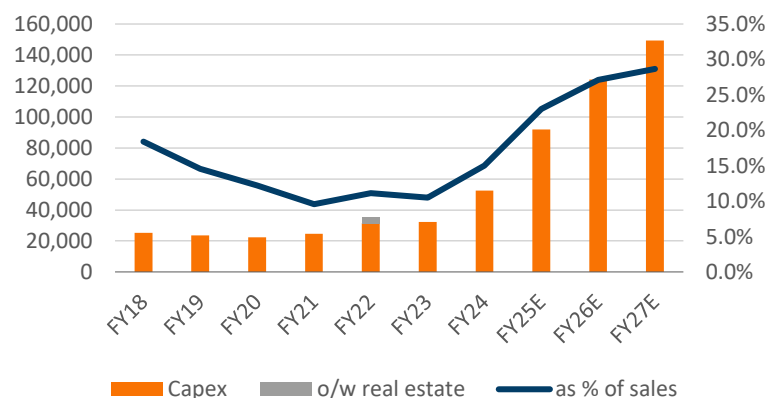
Alphabet has a track record of finding efficiencies, leveraging infrastructure edge.

Efficiency Helps Free Up Capital for Capex

- Capex/sales reaching record level, even as capex has replaced or limited opex cost increases.
- AI is fundamentally applied to *internal* efficiencies, resulting in “smoothed” group EBIT margins of 33.9%** (despite differing business mixes) in each of the three Qs in '25 (excl. one-off Texas and EU fines).

- Replace opex with capex – \$230bn+ '21-'25E** (incl. \$4bn of real estate, reduces rental costs)
- One-time benefit of 110bps to EBIT margins in '23** from extending server lives in D&A

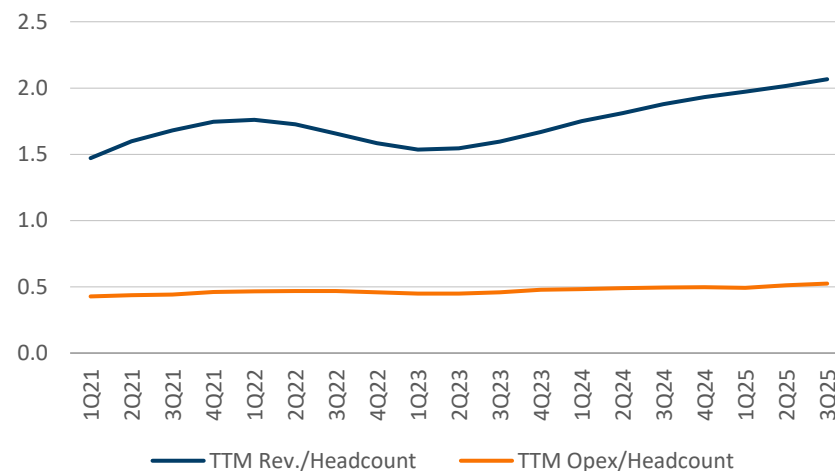
Fig. 7: GOOG's Capex Reaching Record Levels (\$100bn+)



Source: Company data, Arete Research estimates

- Record margins stemmed from increasing sales per employee, while keeping opex under control.**

Fig. 8: Google Raising Headcount, Controlling Opex



Source: Company data, Arete Research

We see further scope for reducing costs in 3P TAC, leverage of content licensing.

EV/EBITDA, P/E Now Closer to Peers

5Y FY2 EV/EBITDA Multiple:

- GOOGL (yellow) now at 18x
- AAPL (purple) at 24x
- META (red) and MSFT (blue) losing some ground

5Y FY2 P/E Multiple:

- GOOGL (yellow) now at 29x, in line with AAPL (purple) at 30.5x and AMZN (grey) at 29x
- Above MSFT (26x)
- Surpassing META (red) recently

What has mostly happened is GOOG erasing the peer group discount.

- We have long argued for Apple to trade at a premium (30x FY2 P/E) due to the *duration* in its business model, ecosystem lock-in, and global brand.
- Google is the #4 global brand, with YouTube at #13.
- On higher estimates, multiple expansion, and more positive perception of Google's position in AI, we raise our target P/E from 25x to 30x and increase our target EV/EBITDA multiple to 18x.
- Applied to our FY27E EPS estimates of \$13.2 (vs. cons. of \$12.7) and EBITDA of \$235bn (vs. \$217bn in cons.), we get an avg. \$380 target price and maintain our Buy rating.
- Given multiple catalysts and momentum behind its AI roadmap and Gemini distribution, **GOOG** remains our top pick among large-cap Internet names.

Table 8: Different Valuation Approaches to Alphabet

Valuation	
30x '24 FY2 P/E + net cash	\$404
18x EV/EBITDA + net cash	\$355
Blended result	\$380

Source: Arete Research estimates.

We are not assuming further material multiple expansion, but do see margins rising from 34% (ex-one-offs) in '25E to 41% by '30E, driven by:

- Operating leverage in Cloud growth
- Rising Subscriptions (esp. of Gemini + YouTube)
- Lower drag from Network and Hardware
- Ongoing “durable” efficiency programmes

Ample scope for “addition by subtraction” – cutting losses, improving focus.

Appendix: Extensive Work in '24-'25

- [*Arete Digital Download: November '25 Edition*](#) – Sections 1 & 6 (Nov. '25)
- [*AI and Big Tech: Hunger Games, Coming Soon*](#) (Oct. '25)
- [*Arete Digital Download: October '25 Edition*](#) – Section 2 (Oct. '25)
- [*Arete Digital Download: September '25 Edition*](#) – Section 2 (Sept. '25)
- [*Alphabet: Network Spin Scenario Increasingly Compelling*](#) (Sept. '25)
- [*Alphabet: Antitrust Ends With Whimper, Not a Bang*](#) (Sept. '25)
- [*Arete Digital Download: August '25 Edition*](#) – Section 1 & 2 (Aug. '25)
- [*Alphabet: Turning the Page*](#) (July '25)
- [*Digital Ads 2Q25: Fragmentation Begins*](#) (July '25)
- [*Arete Digital Download: July '25 Edition*](#) – Section 6 (July '25)
- [*AI Adoption: Big Tech's New OS*](#) (June '25)
- [*Arete Digital Download: June '25 Edition*](#) – Section 1 (June '25)
- [*Alphabet: Perception vs. Reality*](#) (May '25)
- [*Arete Digital Download: May '25 Edition*](#) – Section 2 (May '25)
- [*Alphabet: Life After Antitrust*](#) (April '25)
- [*Arete Digital Download: April '25 Edition*](#) – Sections 2, 3 & 4 (April '25)
- [*Alphabet: Spring Cleaning*](#) (March '25)
- [*Arete Digital Download: March '25 Edition*](#) – Section 4 (March '25)
- [*Social Media: Hunger Game Theory?*](#) (Feb. '25)
- [*Arete Digital Download: February '25 Edition*](#) – Section 5 (Feb. '25)
- [*Alphabet: Time for House-Cleaning*](#) (Dec. '24)
- [*Arete Digital Download: November '24 Edition*](#) – Section 1 (Nov. '24)
- [*Digital Ads 3Q24: Soft-ish Landing*](#) (Oct. '24)
- [*Arete Digital Download: October '24 Edition*](#) – Sections 1 & 2 (Oct. '24)
- [*Alphabet: Cracking the Antitrust Nut*](#) (Sept. '24)
- [*Arete Digital Download: August '24 Edition*](#) – Sections 3 & 4 (Aug. '24)
- [*Alphabet: Winds of Change*](#) (July '24)
- [*Digital Ads 2Q24: Pricing-Led Growth*](#) (July '24)
- [*Arete Digital Download: July '24 Edition*](#) – Sections 1, 4 & 8 (July '24)
- [*Big Tech Capex: The Half-Trillion Dollar Question*](#) (June '24)
- [*Arete Digital Download: June '24 Edition*](#) – Sections 1 & 4 (June '24)
- [*Arete Digital Download: May '24 Edition*](#) – Sections 3 & 4 (May '24)
- [*Alphabet: Flipping the Script*](#) (April '24)
- [*Digital Ads: Tougher Comps Loom Post-1Q24*](#) (April '24)
- [*Arete Digital Download: April '24 Edition*](#) – Sections 5 & 6 (April '24)
- [*Arete Digital Download: March '24 Edition*](#) – Section 1 (March '24)
- [*Privacy Sandbox, iOS and Beyond: Rewiring Digital Ads Markets*](#) (Feb. '24)
- [*Arete Digital Download: February '24 Edition*](#) – Section 1 (Feb. '24)
- [*Internet and Digital Consumer: Outlook for '24: A Tale of Two Cities*](#) (Jan. '24)

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Required Disclosures

Alphabet (Buy), Target Price: \$380 (FY'26E)

Valuation Methodology: Our FY26E price target of \$380 is based on a target 30x '26E P/E + net cash (fair value of \$404), a target 18x '26E EV/EBITDA + net cash (fair value of \$355).

Investment Risks (to our Rating and Target Price):

1. Advertising demand may falter under macroeconomic pressures and tough comps.
2. Antitrust trials could lead to severe restrictions on Google's business.
3. Google PlayStore may be forced to reduce take-rates or offer 3P billing options.
4. Google investments in AI fail to develop into monetised services or products.
5. YouTube (TV and Music) could fail to take more market share or incur higher content costs.
6. Alphabet's many Other Bets and early-stage investments could never cease losses.
7. Google Cloud could see growth rates slow, or prove to have grossly overpaid for Wiz (\$32bn).
8. Competitors could develop alternatives to Google Search, undermining its highest-margin business.
9. Alphabet management could fail to take action around costs and a politicised organisation.

Primary Analyst(s) Coverage Group: Adyen, Alibaba, Alphabet, Amazon.com, Apple, AppLovin, Baidu, Block, CD Projekt, Chime, Criteo, eBay, Electronic Arts, Ericsson, Etsy, Global-e Online, JD.com, LiveRamp, Lyft, Meituan, Meta Platforms, Nokia, Paypal, Pinterest, Roblox, Roku, Samsung Electronics, Sea Limited, Shopify, Snap, Sony, Spotify, Take-Two Interactive, Tencent, The Trade Desk, Uber Technologies, Unity Software, Xiaomi, Zalando.

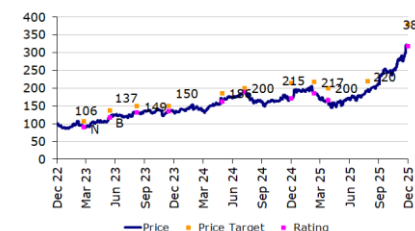
Potential Conflicts: None.

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Distribution of Ratings – As of 30 September 2025, 59.0% of stocks covered were rated Buy, 10.1% Sell and 30.9% Neutral.

GOOG Price Chart



Source: Arete Research estimates.

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